

TECHNOLOGICAL UNIVERSITY (THANLYIN)
DEPARTMENT OF INFORMATION TECHNOLOGY ENGINEERING

JOURNAL READING RECOMMENDATION SYSTEM

BY

MA SWE ZIN OO

VI IT - 5

GRADUATION THESIS

OCTOBER, 2025

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OCTOBER, 2025

THANLYIN

TECHNOLOGICAL UNIVERSITY (THANLYIN)
DEPARTMENT OF INFORMATION TECHNOLOGY ENGINEERING

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ABSTRACT

The rapid growth of online academic journals has created a need for efficient and personalized tools that help students and researchers access relevant literature quickly. Manual searching is time-consuming and often ineffective, hindering research productivity. This mini thesis addresses this issue by developing a Journal Reading and Recommendation System that provides personalized suggestions based on user behavior and preferences. The system analyzes reading history and tag preferences to generate recommendations using Content-Based Filtering, ensuring that suggested journals align with individual interests. Users can track their reading history, manage settings, check subscription history, and view recommended journals, while administrators can manage journals, categories, and tags. Implementation uses Firebase Firestore for data storage, Supabase Storage for journal files, and a real-time database for background email processing, ensuring reliable notifications. The mobile application is developed with Dart and Flutter, incorporating real-time filtering, chart visualization, and platform-aware features to enhance user experience. Software design patterns such as Singleton, Observer, and Strategy are applied to improve maintainability, scalability, and code organization. Results show that combining content-based filtering with structured data analysis produces relevant, diverse, and personalized recommendations. Real-time filtering, chart visualization, and cloud-based file management improve engagement and accessibility. The system effectively reduces the time users spend searching for journals while delivering recommendations tailored to individual preferences. This research provides a foundation for integrating intelligent analytics, including adaptive recommendation strategies, real-time notifications, and personalized content prioritization. Ultimately, the system enhances academic efficiency, resource utilization, and user satisfaction by making journal discovery faster, more accurate, and highly personalized.

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